



Intro to Git and GitHub



Center for Research Data
& Digital Scholarship

**How do you plan on
using Git and GitHub?**

Git vs GitHub

- Git: version control system
 - the actual software
- GitHub
 - cloud-based storage website



What is Git?

- Git is version control software, created by Linus Torvalds, the same person who created the Linux operating system.
- Git monitors files on your computer and tracks changes made to them over time
- Uses the command line

Why is it called “Git”?

- Linus Torvalds: "I'm an egotistical bastard, and I name all my projects after myself. First 'Linux', now 'Git'".
- Git: “A silly, incompetent, stupid or annoying person (usually a man).” (Wikitionary)

git / README 



Linus Torvalds Initial revision of "git", the information manager from hell

Code

Blame

168 lines (135 loc) · 8.2 KB

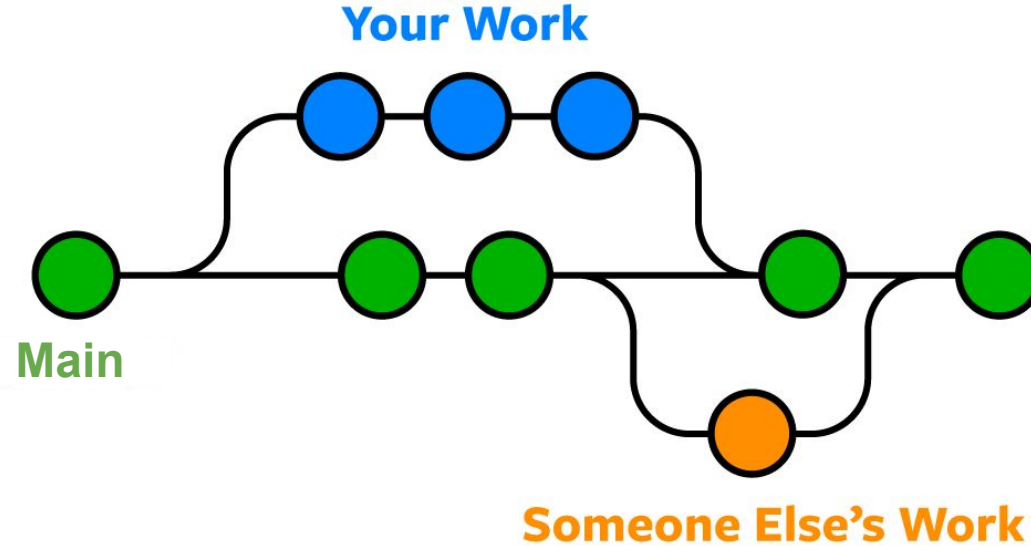
```
1
2     GIT - the stupid content tracker
3
4     "git" can mean anything, depending on your mood.
5
6     - random three-letter combination that is pronounceable, and not
7       actually used by any common UNIX command. The fact that it is a
8       mispronunciation of "get" may or may not be relevant.
9     - stupid. contemptible and despicable. simple. Take your pick from the
10      dictionary of slang.
11     - "global information tracker": you're in a good mood, and it actually
12       works for you. Angels sing, and a light suddenly fills the room.
13     - "goddamn idiotic truckload of sh*t": when it breaks
14
```


What is version control?

- Version control is the practice of tracking and managing changes to files.
 - Revert to various states of files
 - Modify items without harming the original copy
- Not limited to code
 - Can be used for documents, images, etc.

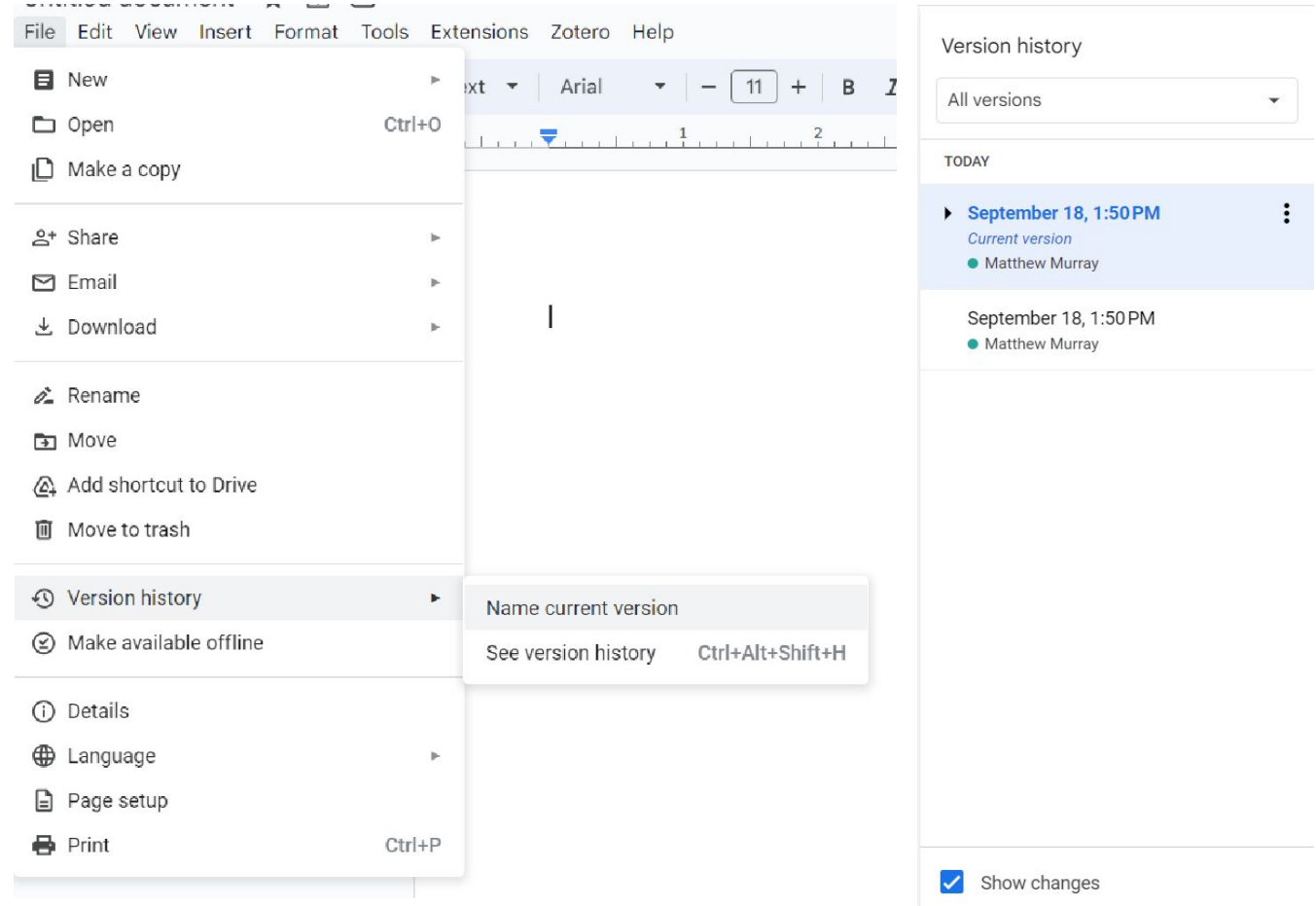
Benefits of version control

- Using version control provides
 - Clear tracking of the repo's history
 - Management and view of different branches
 - Collaboration through merging of branches

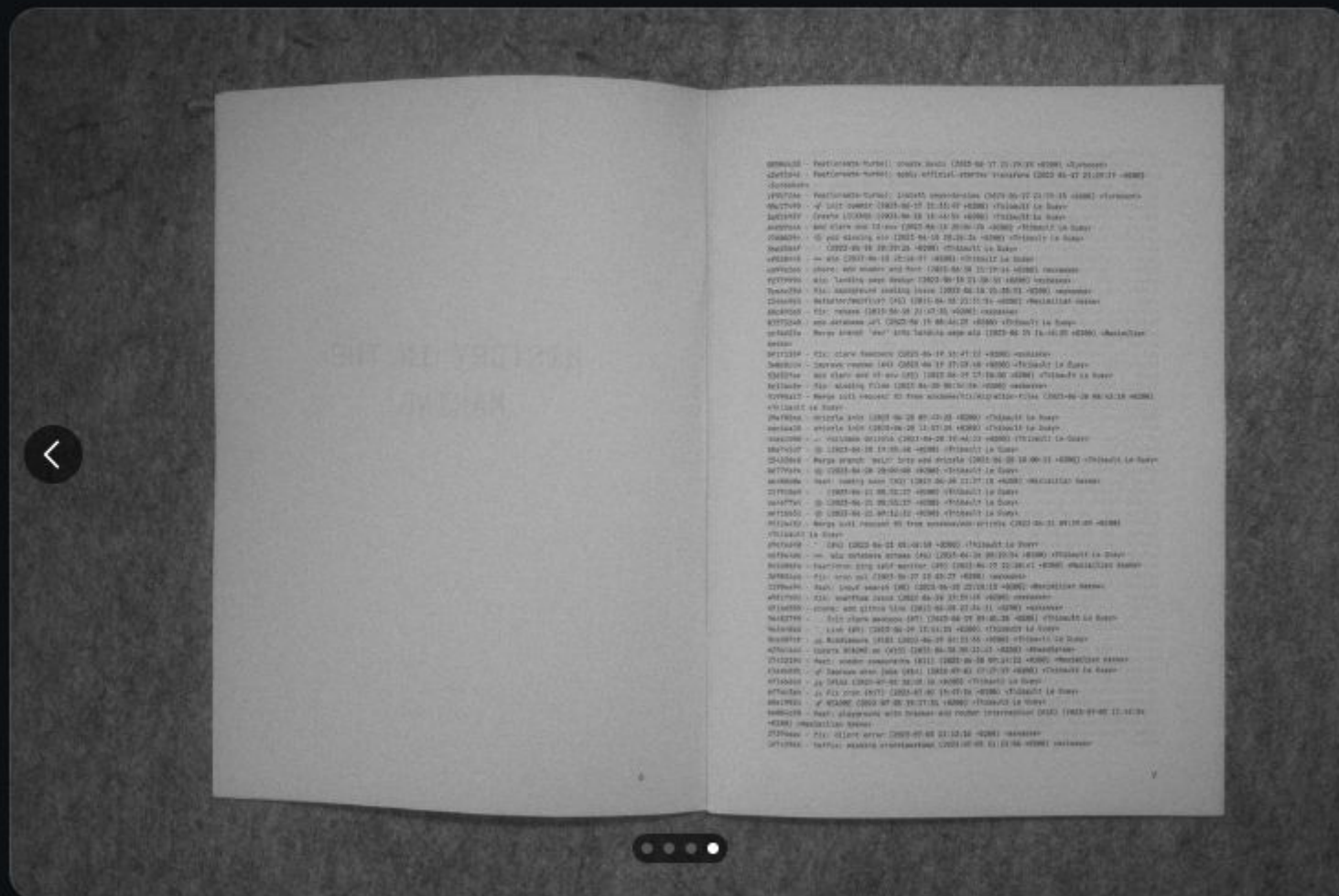


What is version control?

- Google Docs includes "Version History"
- This allows you to see what changes were made, when those changes happened, and who made them
- You can also revert to a previous version of your file



We made a zine from our Git history



We are an open-source project, so we decided to print our Git history to celebrate our 2nd anniversary.

https://www.reddit.com/r/zines/comments/1miy3bs/we_made_a_zine_from_our_git_history/

What is GitHub?

- GitHub is a Microsoft subsidiary that offers cloud hosting for Git repositories
- Provides a GUI (Graphical User Interface) for Git
 - Most (but not all) features are available in the GitHub Desktop client
- Allows for easy collaboration and sharing
 - Issue queues for bugs and features, pull requests, and more
- GitHub basic is free (has limits)
 - Hosts both open and private repositories
- GitHub Enterprise (free for CU affiliates)
 - Includes cloud-based development environments
 - <https://oit.colorado.edu/services/business-services/github-enterprise>

What Can I Use Git and GitHub For?

Git

- Track and manage changes of files
- Document changes
- Collaborate with others

GitHub

- Off-site backup for Git repositories
- Contribute to Open Source projects
- Communicate with team members
- Communicate with your users
- Join communities
- Bug tracking
- Website hosting

<> Code ▾

[cu-boulder-crdds.github.io/Open-Access...](https://cu-boulder-crdds.github.io/Open-Access-Data)

Report repository

Create a new release

Packages

 ellery-galvin Add ID to ring image and update CSS with target 	fbfd88b · 3 weeks ago	 37 Commits
 .gitignore	Adding .gitignore. Resizing images.	3 weeks ago
 CU-Scholar.png	Images and styles.	last month
 ILL.png	Images and styles.	last month
 LICENSE	Create LICENSE	3 weeks ago
 OA-Logo.jpeg	Images and styles.	last month
 OA_Week_2025.html	Add ID to ring image and update CSS with target	3 weeks ago
 OneSearch.png	Updating files, adding image	2 months ago
 README.md	Update README.md	3 weeks ago
 UnderGrad-01.png	Fixing issues.	3 weeks ago
 University_Libraries_left.png	Updating logo.	3 weeks ago
 grad-01.png	Fixing issues.	3 weeks ago



Try to navigate academic life without spending your takeout food budget!

Learn about:

- Access to scholarly articles
- Publishing scholarly articles
- Impact and scholarly monographs

Choose your character:

Undergrad Student

Grad Student

Faculty Member

Alternatives to...

- Git (version control)
 - Apache Subversion (SVN)
 - Mercurial SCM
 - CVS (Concurrent Versions System)
- GitHub (hosting)
 - Codeberg
 - GitLab
 - Forgejo
 - Bitbucket



When **NOT** to use GitHub

- When you are looking for long-term preservation
 - There's no guarantee Microsoft will keep GitHub around forever
 - Lots of source-code-hosting platforms no longer exist
 - Thousands of URLs in research articles no longer work as they point to code hosted by defunct services
- When you want your code to be cited
 - [Zenodo](#) is an Open Access repository that can be linked to GitHub
 - Allows you to archive a specific release of public GitHub projects
 - Creates a DOI for the archive that you can use in citations

Zenodo.org will be unavailable for 2 hours on September 29th from 06:00-08:00 UTC. See [announcement](#).

September 2, 2023

Software

Open Access

PyGMT: A Python interface for the Generic Mapping Tools

 Tian, Dongdong;  Uieda, Leonardo;  Leong, Wei Ji;  Schlitzer, William;  Fröhlich, Yvonne;  Grund, Michael;  Jones, Max;  Toney, Liam;  Yao, Jiayuan;  Magen, Yohai;  Jing-Hui, Tong;  Materna, Kathryn;  Belem, Andre;  Newton, Tyler;  Anant, Abhishek;  Ziebarth, Malte;  Quinn, Jamie;  Wessel, Paul

PyGMT is a library for processing geospatial and geophysical data and making publication quality maps and figures. It provides a Pythonic interface for the [Generic Mapping Tools \(GMT\)](#), a command-line program widely used in the Earth Sciences.

The development of PyGMT has been supported by NSF grants OCE-1558403 and EAR-1948603.

Preview

 baseline-images.zip

baseline-images

- test_basemap.png 6.2 kB
- test_basemap_compass.png 71.2 kB
- test_basemap_loglog.png 24.5 kB
- test_basemap_map_scale.png 31.0 kB
- test_basemap_polar.png 31.9 kB

18,263

 views

1,446

 downloads

[See more details...](#)

Available in

GitHub

Indexed in

OpenAIRE

Publication date:

Zenodo.org will be unavailable for 2 hours on September 2, 2023

September 2, 2023

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Preview

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baseline-images

- test_basemap.png
- test_basemap_compass.png
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- test_basemap_map_scale.png
- test_basemap_polar.png

Versions

Version v0.10.0

Sep 2, 2023

10.5281/zenodo.8303186

Version v0.9.0

Mar 31, 2023

10.5281/zenodo.7772533

Version v0.8.0

Dec 30, 2022

10.5281/zenodo.7481934

Version v0.7.0

Jul 1, 2022

10.5281/zenodo.6702566

Version v0.6.1

Apr 11, 2022

10.5281/zenodo.6426493

[View all 17 versions](#)

Cite all versions? You can cite all versions by using the DOI [10.5281/zenodo.3781524](#). This DOI represents all versions, and will always resolve to the latest one. [Read more.](#)

24.0 KB

31.0 kB

31.9 kB

18,263

 views

1,446

 downloads

[See more details...](#)

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GitHub

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OpenAIRE

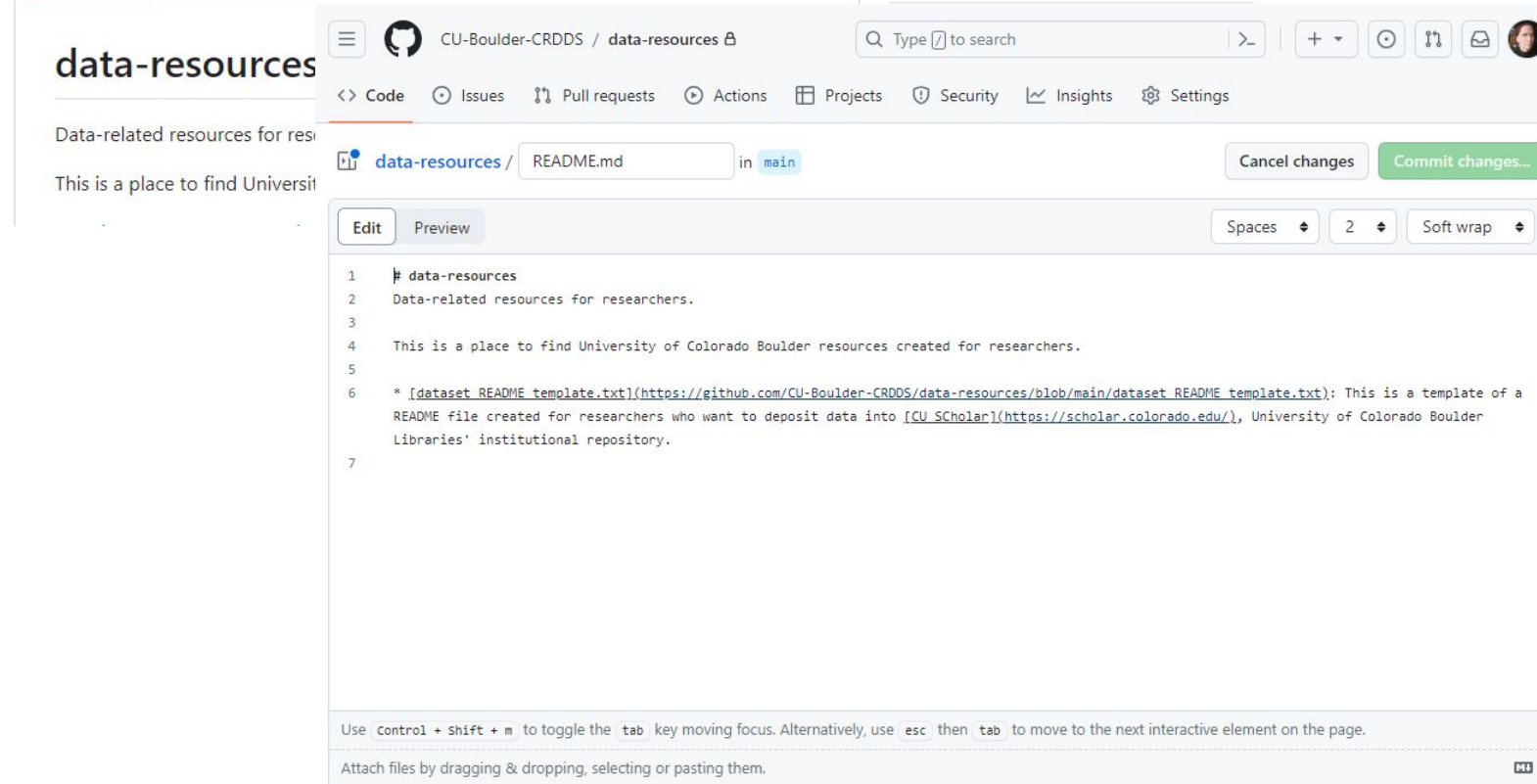
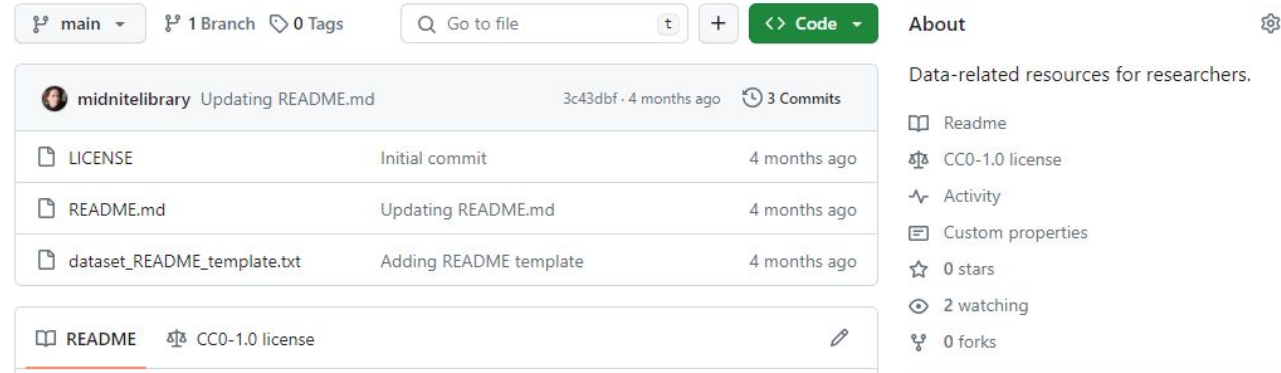
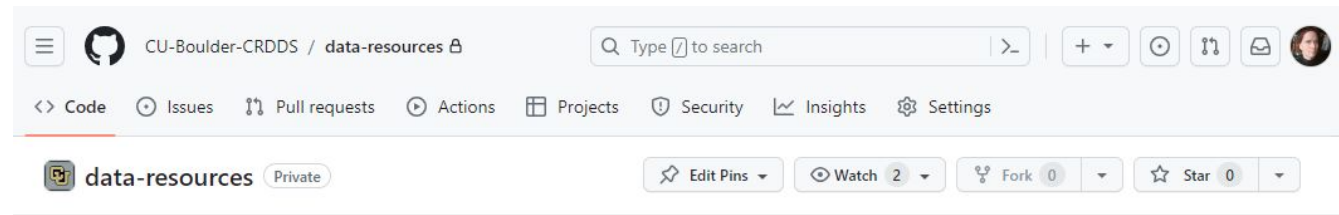
Publication date:

Ways to interact with GitHub

- The website
- GitHub Desktop
- Using Git and the command line

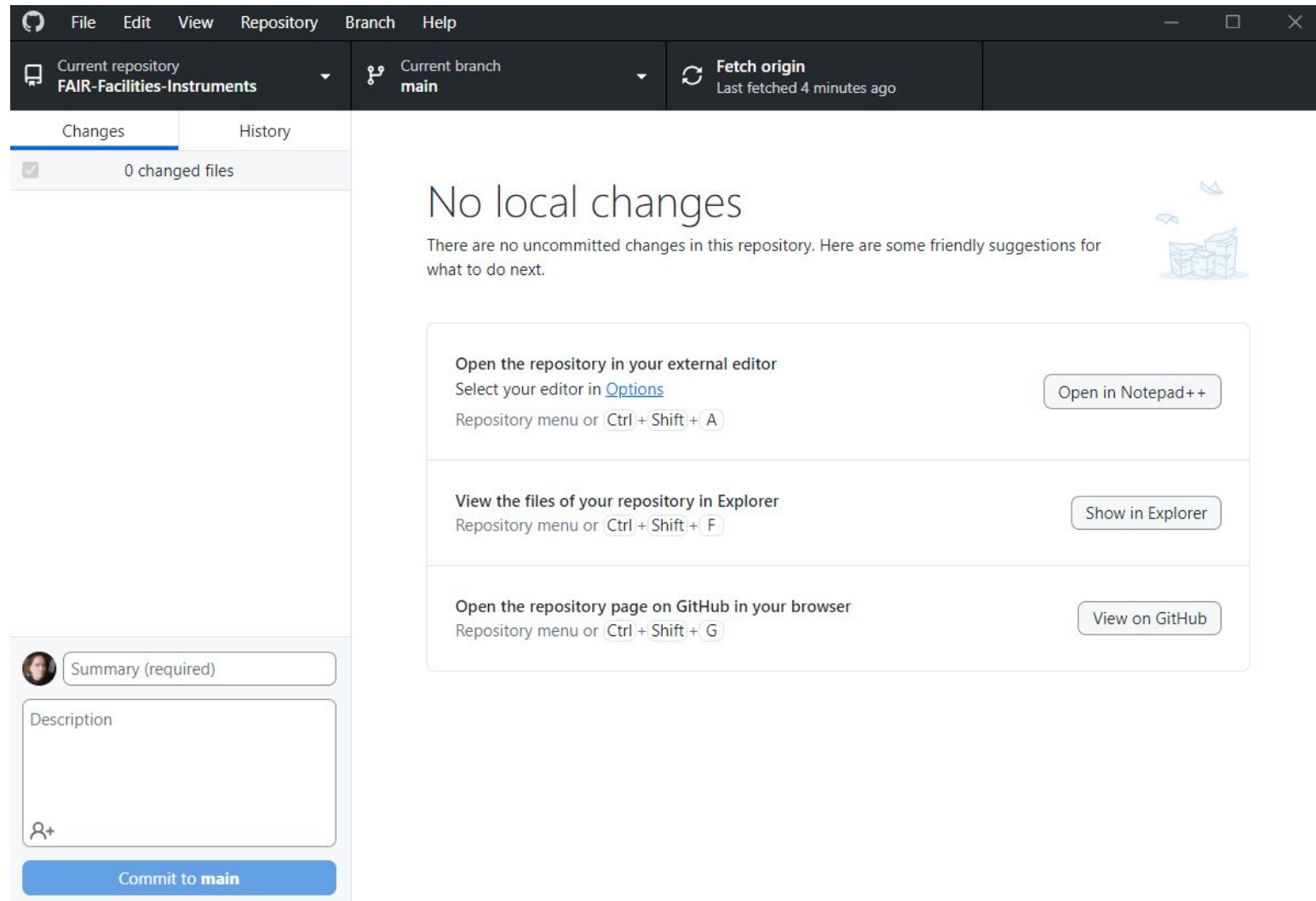
Website

- Create and manage projects
- Upload and download files
- Write documentation



GitHub Desktop

- Create, clone, and fork projects
- Commit changes and submit contributions
- Many people who use Git frequently prefer to use the command line
- Other graphical Git interfaces exist



Be careful where you put your GitHub directory

University of Colorado Boulder OneDrive

Search

+ Add new

Share Copy link Download Export to Excel Automate

Sort Details

Matthew Murray

My files > UCB > Documents > GitHub

Name	Modified	Modified By	File size	Sharing	Activity
.git	September 12, ...	Matthew Murray	5 items	Private	
.github	April 29, 2024	Matthew Murray	2 items	Private	

- Microsoft OneDrive uploads a lot of stuff to the cloud, which you don't always want

Getting Started with Git

Definitions

- Trunk/Master: Main version of a repository
- Clone: A copy of a repository that is stored locally instead of online
- Branch: A parallel version of a repository that allows modifications to be made without affecting the main file
- Fork: A branch that is not intended to be merged back into the main file
- Pull/Fetch: Getting the latest version of a repository
- Merge: Combining changes from different branches/versions of files
- Commit: A saved snapshot of your project and any changes you have made
- Push: Sending your local changes to an online repository

Things to Remember about the Git Workflow

- Git isn't an automatic process (it's not like autosave)
- In order to use it, you need to integrate Git into your daily workflow
- You set it up for a project, on a directory on your hard drive, and from that point forward Git will start watching for any changes in that directory
- You do your work as normal, using all the same software and tools you're used to using
- At any point, you can ask Git to tell you what changes it has detected and it will show you a detailed report of every addition and deletion that it has detected
- When you want to you can make a commit and create a snapshot of your project

Things to Remember about the Git Workflow

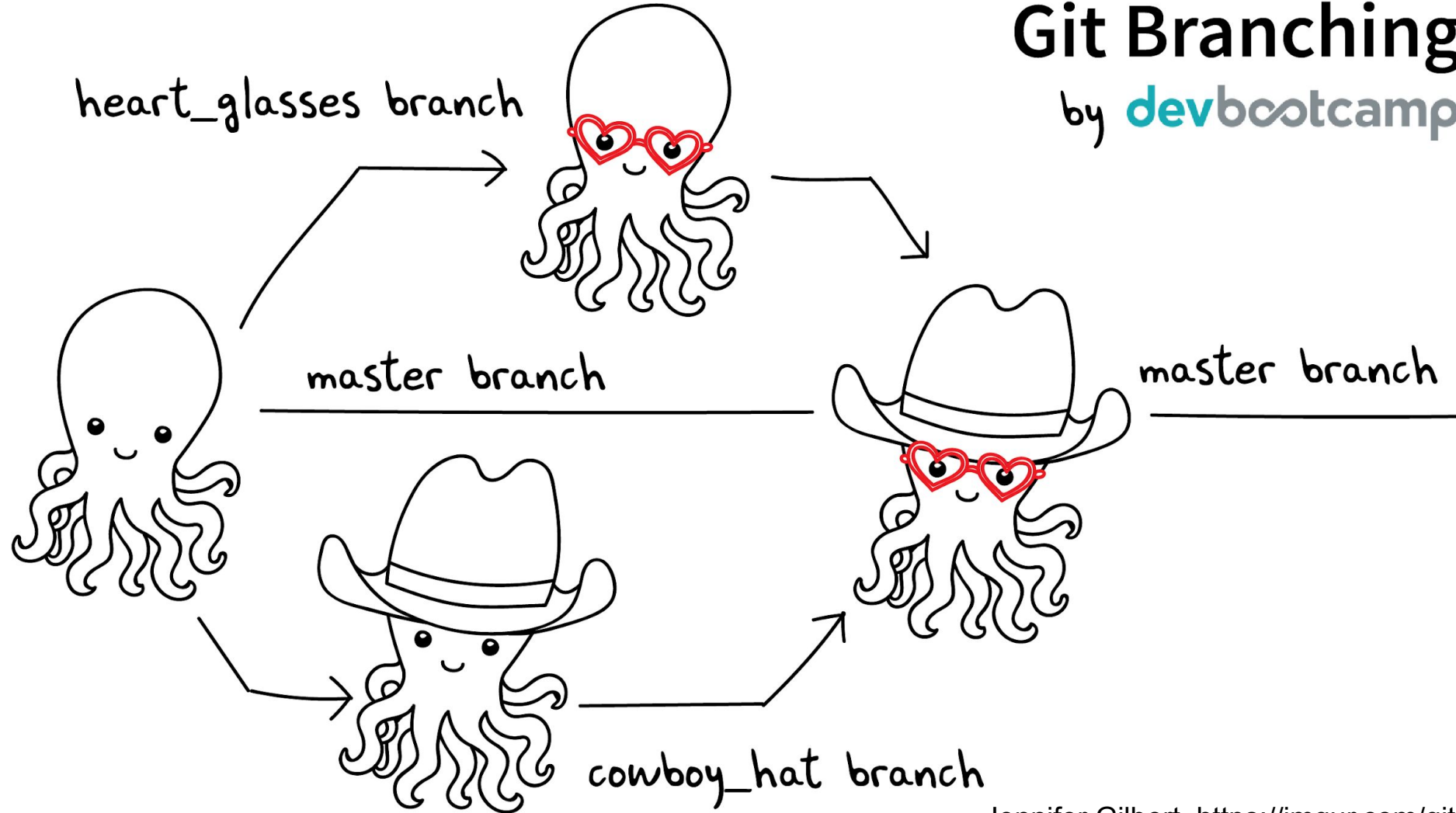
- Once you've finished working on something, it's time to make a commit, which you can think of as a snapshot of the state of your project. Each commit records all the changes that happened to get it from the previous version of your project to the current one.
- When you make a commit, you select some (or all) of the changes that git has detected in your directory, and you mark them for inclusion in your commit. Then, you provide a message to describe the changes that you've made and create your commit.
- The commit is added to a journal of sorts, where you can see all the various commits that you've recorded for your project.
- Then, at any point in the future, we can examine any of the commits, look at the files that changed, and the message that was created for the commit. If you decide that you need to, you can “rewind” your project to the state that's described by the commit.

Git Workflow - Branches

- Branches allow you to create multiple versions of your project, and switch between them seamlessly.
- Git is often described as a “distributed” version control system. When git is described as “distributed” it means that it’s designed to work with many copies of the same repository in many different locations, different computers or servers. Those individual copies can then speak to each other, to exchange information and share changes between each of them.
- If you’re working on your own, this might not seem like a useful feature, but it becomes really valuable when you’re working with other people.
- At any point, each person can start working independently on their own copy of code and use the branching and merging functionality to do their work.
- Each branch lives on a different computer entirely, but the changes can be pushed back into the main copy.

Branching & Merging

Git Branching
by **dev**bootcamp



Areas of Git Workflow

Working Area

- Items that you are currently working on
- Are not tracked by Git!
- Exists locally
- Created by using “git init”

Staging Area

- When Git starts tracking and saving your work
- Exists locally
- Items are added to this area by using “git add”

Snapshot Area

- All staged items are captured
- Version of the repo
- Exists locally
- Items are added to this area by using “git commit”

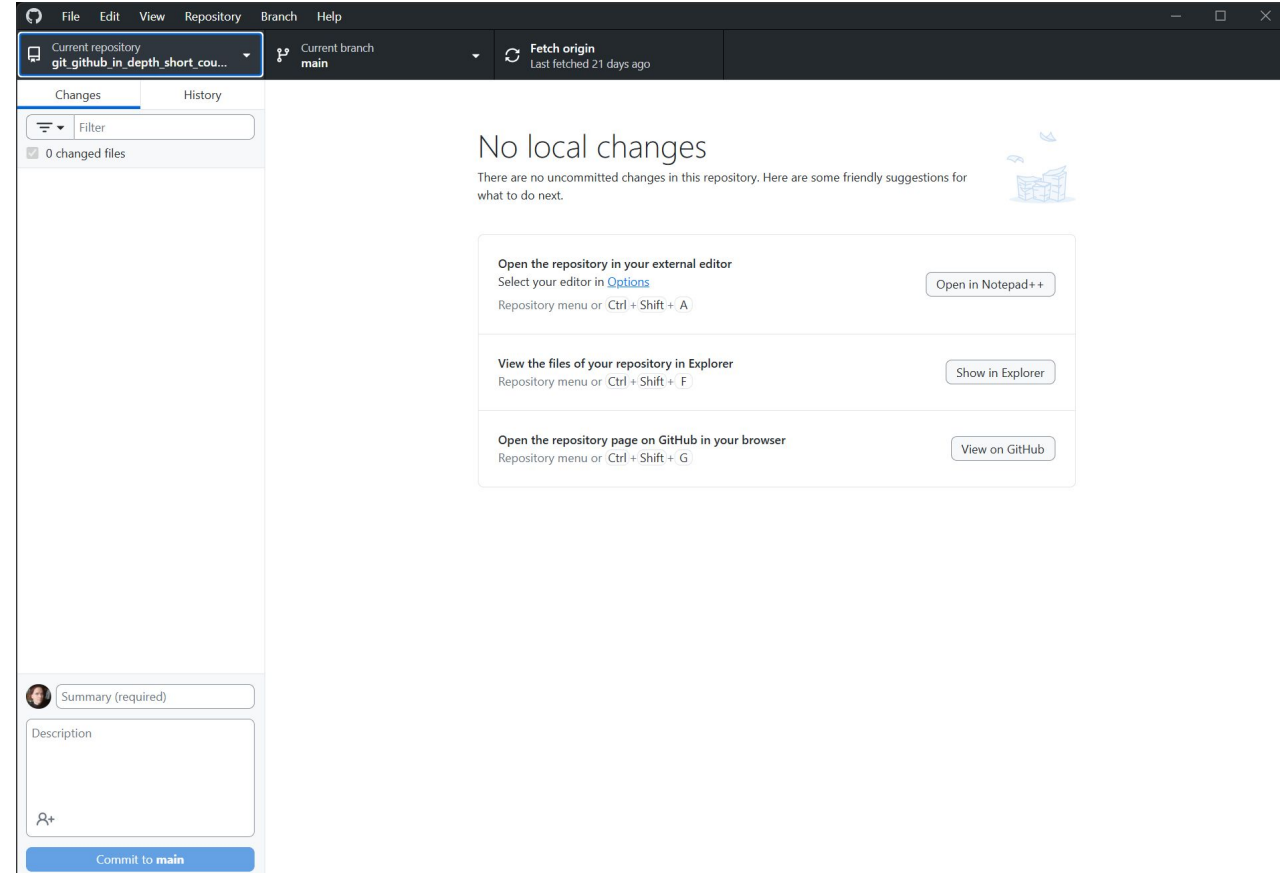
GitHub

- Exists locally and on GitHub!
- Items are added to this area using “git push”

Interacting with Git & GitHub

GitHub Desktop

- Easiest way to get started
- Offers a very streamlined experience
- Simplicity comes at the expense of power and flexibility. A lot of advanced Git features aren't available in the GitHub Desktop client.
- GitHub desktop only works with local Git and GitHub. If your team is using GitLab, BitBucket, or some other cloud repository hosting service, this client will not speak to those.



The Command Line

- The most powerful, and the best way to really learn the ins and outs of Git.
- Has the steepest learning curve. The command line interface requires you to learn and memorize all the available commands, the workflow, etc.
 - [Cheat sheets](#) are available!
- Not a good option for beginners unless you're the kind of person that's already used to working on the command line.

Getting Git on your local machine

- Many systems have Git installed; however, you may need to download it on your local machine:
- See <https://git-scm.com/book/en/v2/Getting-Started-Installing-Git> for more information on installing Git
- You don't need to do this if you're just using the graphic version of GitHub

Linking Git to GitHub

- Once downloaded, we can then configure Git with our GitHub username and email via the command line. This allows us to interact with our GitHub more easily.

- First, set your username. For example, if my GitHub username is gh-user, then I would do the following:

```
$ git config --global user.name "gh-user"
```

- Now, set your email. For example, if my email for GitHub is gh-user@gmail.com, then I would do the following:

```
$ git config --global user.email "gh-user@gmail.com"
```

- Confirm Git has been configured (should show your entered info)

```
$ git config --list
```


Personal Access Token

- Allows you to verify your identity with GitHub
- For more information, see <https://docs.github.com/en/authentication/keeping-your-account-and-data-secure/managing-your-personal-access-tokens>
- Downside is that you need to enter your username and then the Personal Access Token as your password for events such as:
 - Any interaction with a private repo
 - Pushing to a public repo
- There are ways to store your username and token, but these require third-party software

SSH Keys

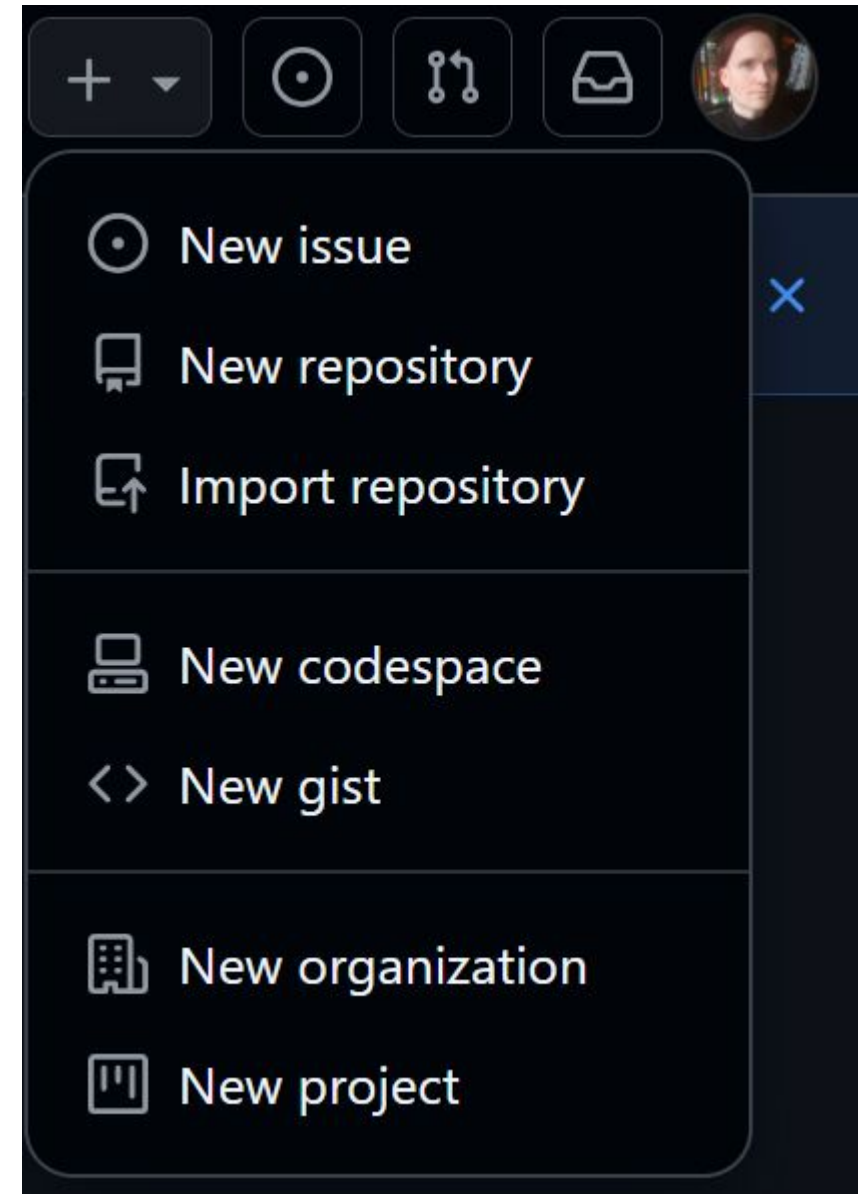
- Alternative way to verify your identity with GitHub
- For more information see:
<https://docs.github.com/en/authentication/connecting-to-github-with-ssh/generating-a-new-ssh-key-and-adding-it-to-the-ssh-agent>
 - Be sure to select the proper Operating System when using the link
- Setup is more involved, but makes it so that you never have to enter your username and token when interacting with a private repo or pushing to a public repo

Setting up a new GitHub repo

New GitHub Repos

When you first create a repo locally, you will need to setup a new repository on GitHub too

- Go to <https://github.com>
- Sign in
- Click on “Create New Repository” or just “New”



Creating a repo

- Choose an “owner”
- Add a description
- Choose visibility
- Add a README
- Add a .gitignore
- Add a license

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *

Choose an owner ▾

Repository name *

Great repository names are short and memorable. How about **turbo-octo-adventure**?

Description

0 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Public ▾

Start with a template

Templates pre-configure your repository with files.

No template ▾

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Off ☐

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore ▾

Add license

Licenses explain how others can use your code. [About licenses](#)

No license ▾

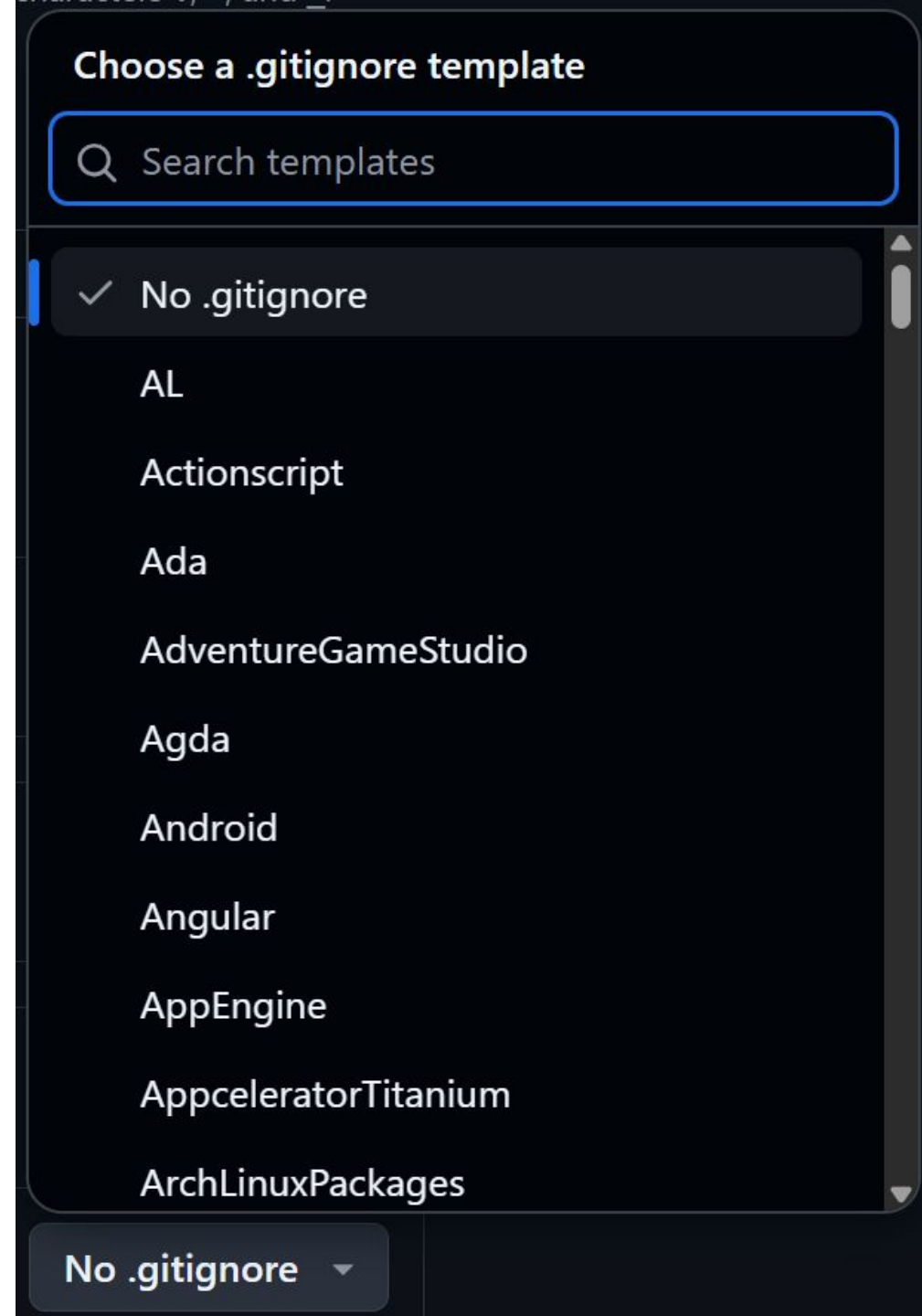
Create repository

README Files & Documentation

- Include documentation with your project in GitHub so that others (and you) know what your project is and how it should be used.
- A README.md (markdown) file can be included in your GitHub project and will display on the front page
- What to include in a README:
 - What your project does
 - How people can use it
 - Who you are and how to contact you
 - License information
- Lots of examples and templates available
- We can provide feedback on documentation

.gitignore

- Specifies what items (files, directories, etc.) should never be tracked by Git
- Use for information that needs to be kept private or is irrelevant to a project
- Use a template or create your own



Example .gitignore

- [CHAFF](#) (Concealed, Hidden, And Forgotten Files)
- .DS_Store files are created automatically by MacOS computers and are not visible to those using that OS
- These files are used to keep track of metadata used by Finder—such as display options or icon positions—and Spotlight's search features

```
1
2 # DS store created on a mac
3 .DS_Store
4 ._DS_Store
5 **/.DS_Store
6 **/._DS_Store
7
8 # PowerPoint temporary
9 ~$*.ppt*
10
```


Licenses

- A license is a legal permission that specifies what you are (and are not) allowed to do a work
- Applies to how you can use, modify, and distribute the work
- Apply to creative work, software, datasets, intellectual property, and more
- Consider how permissive do you want to be
- For code, check out the website “Choose an open source license”
<https://choosealicense.com/>
- Feel free to reach out and ask us
- Note: I am not a lawyer

Choose a license

✓ No license

Apache License 2.0

GNU General Public License v3.0

MIT License

BSD 2-Clause "Simplified" License

BSD 3-Clause "New" or "Revised" License

Boost Software License 1.0

Creative Commons Zero v1.0 Universal

Eclipse Public License 2.0

GNU Affero General Public License v3.0

No license ▾

choosealicense.com

- Has more licenses than the ones available by default on GitHub
- Use this option to add it to your GitHub repo

[Home](#) / [Licenses](#)

Creative Commons Attribution Share Alike 4.0 International

[Copy license text to clipboard](#)

Similar to [CC-BY-4.0](#) but requires derivatives be distributed under the same or a similar, [compatible](#) license. Frequently used for media assets and educational materials. A previous version is the default license for Wikipedia and other Wikimedia projects. Not recommended for software.

Permissions

- Commercial use
- Distribution
- Modification
- Private use

Conditions

- License and copyright notice
- Same license
- State changes

Limitations

- Liability
- Patent use
- Trademark use
- Warranty

Suggest this license

Make a pull request to suggest this license for a project that is [not licensed](#). Please be polite: see if a license has already been suggested, try to suggest a license fitting for the project's [community](#), and keep your communication with project maintainers friendly.

Enter GitHub repository URL

How to apply this license

Create a text file (typically named LICENSE or LICENSE.txt) in the root of your source code and copy the text of the license into the file. It is also acceptable to solely supply a link to a copy of the license, usually to the [canonical URL for the license](#).

Optional steps

Add [CC-BY-SA-4.0](#) to your project's package description, if applicable (e.g., [Node.js](#), [Ruby](#), and [Rust](#)). This will ensure the license is displayed in package directories.

[Source](#)

Who's using this license?

[Flight rules for Git](#)
[Material Design Iconic Font](#)
[OWASP MSTG](#)

Attribution-ShareAlike 4.0 International

=====

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 **RDM-Escape-Room** Public

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

<> Code

 midnritelibrary	Create .gitignore	2338fc1 · now	2 Commits
	.gitignore	Create .gitignore	now
	LICENSE	Initial commit	1 minute ago
	README.md	Initial commit	1 minute ago

README

CC0-1.0 license

RDM-Escape-Room

About

No description, website, or topics provided.

Readme

CC0-1.0 license

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

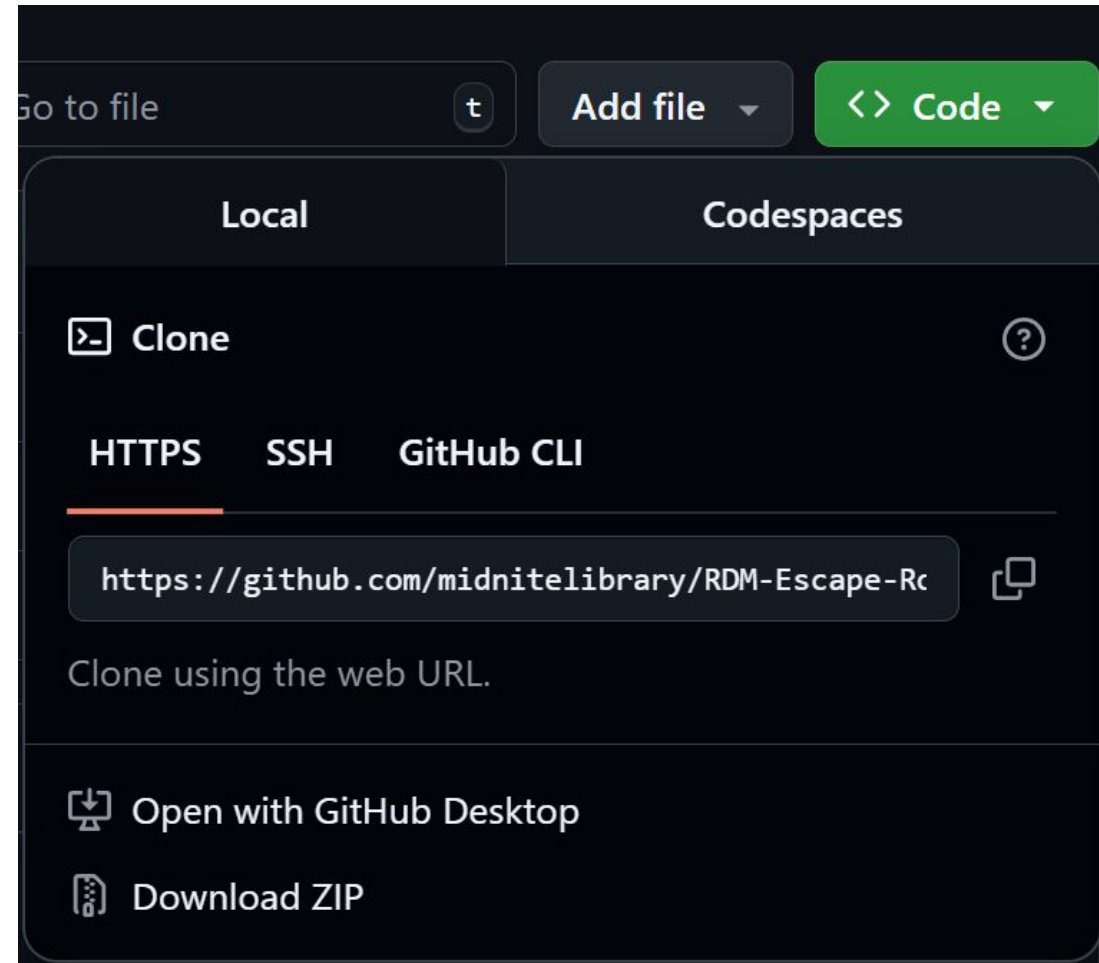
Packages

No packages published

[Publish your first package](#)

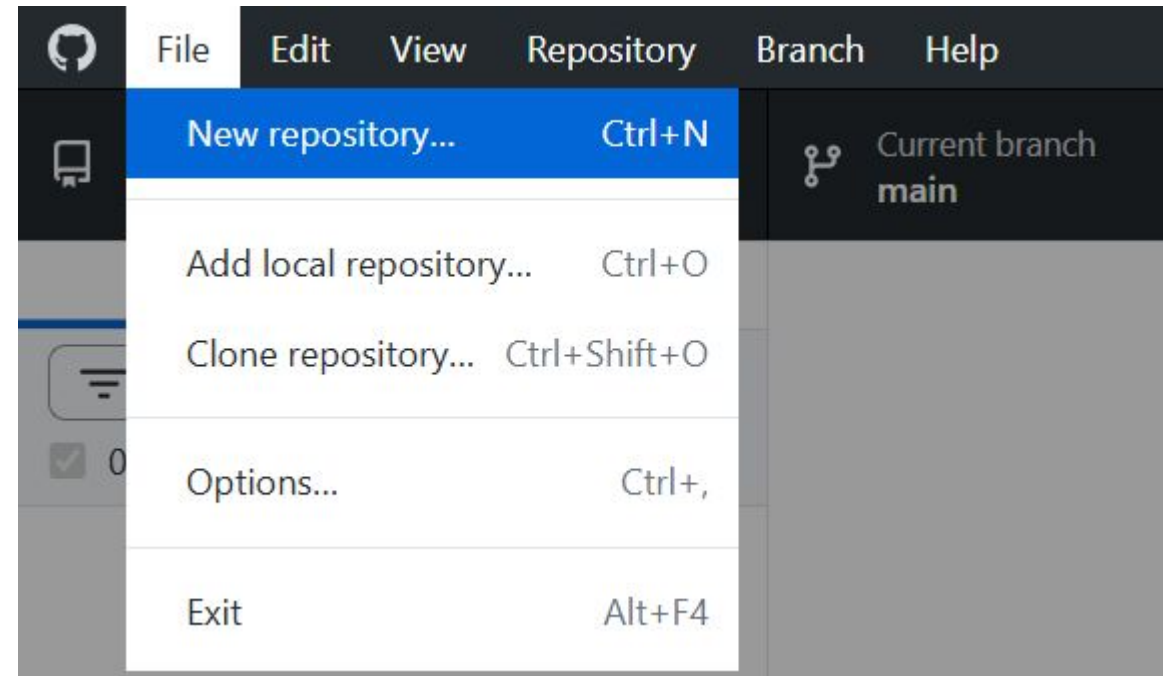
Clone a Repo to Your Computer

- On a GitHub repo click on “Code”
- Copy the URL (or use “Open with GitHub Desktop”)



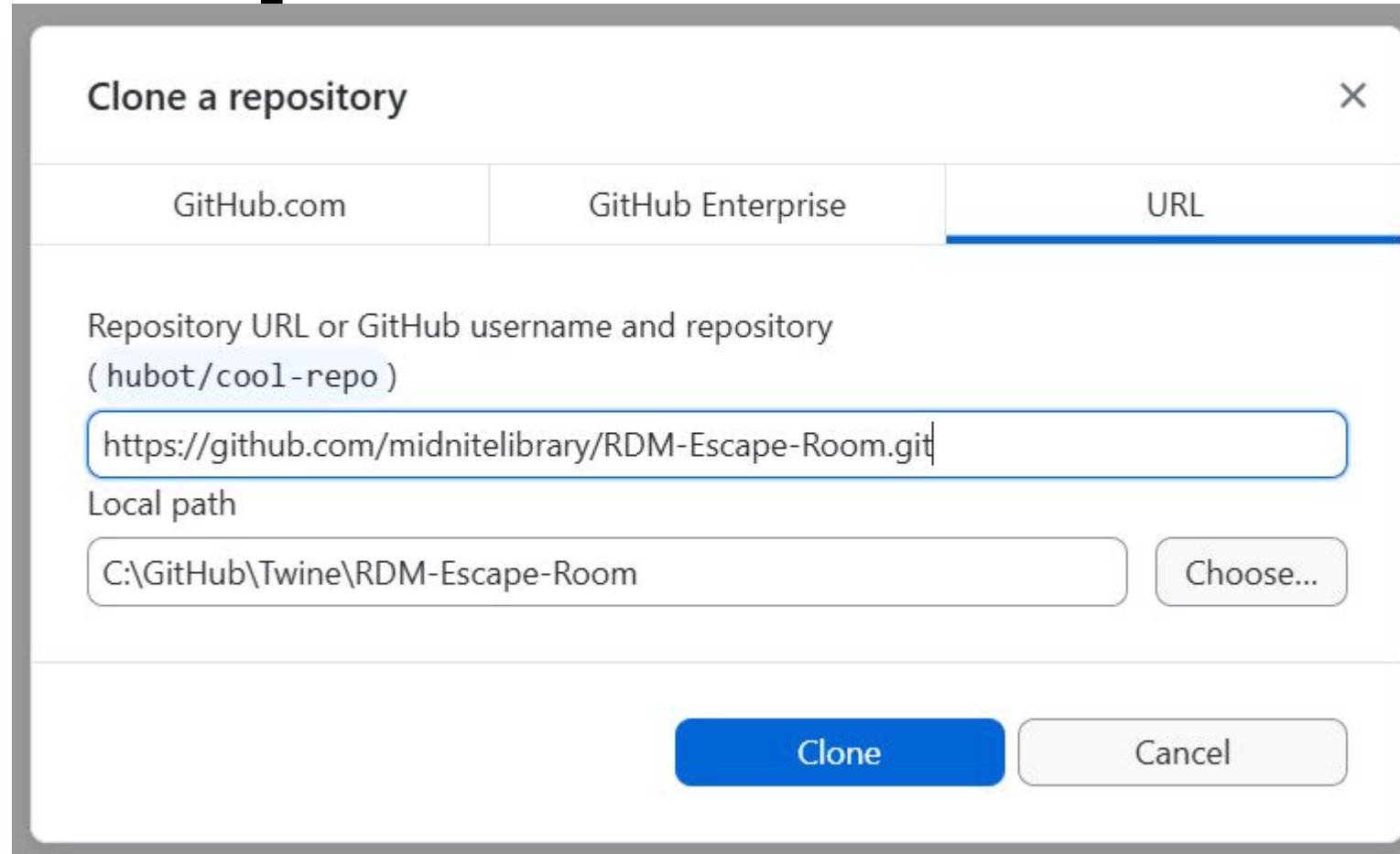
In GitHub Desktop

- Click “File”
- Select “Clone repository”



In GitHub Desktop

- Select “URL”
- Paste in the URL
- Click “Clone”



The screenshot shows the 'Clone a repository' dialog box in GitHub Desktop. It has a title bar with a close button (X). Below the title bar are three tabs: 'GitHub.com', 'GitHub Enterprise', and 'URL'. The 'URL' tab is selected and highlighted with a blue underline. Below the tabs, there is a text input field labeled 'Repository URL or GitHub username and repository (hubot/cool-repo)' containing the URL 'https://github.com/midnritelibrary/RDM-Escape-Room.git'. Below this is a 'Local path' section with a text input field containing 'C:\GitHub\Twine\RDM-Escape-Room' and a 'Choose...' button. At the bottom right are two buttons: 'Clone' (blue) and 'Cancel' (light gray).

Clone a repository

GitHub.com GitHub Enterprise **URL**

Repository URL or GitHub username and repository
(hubot/cool-repo)

https://github.com/midnritelibrary/RDM-Escape-Room.git

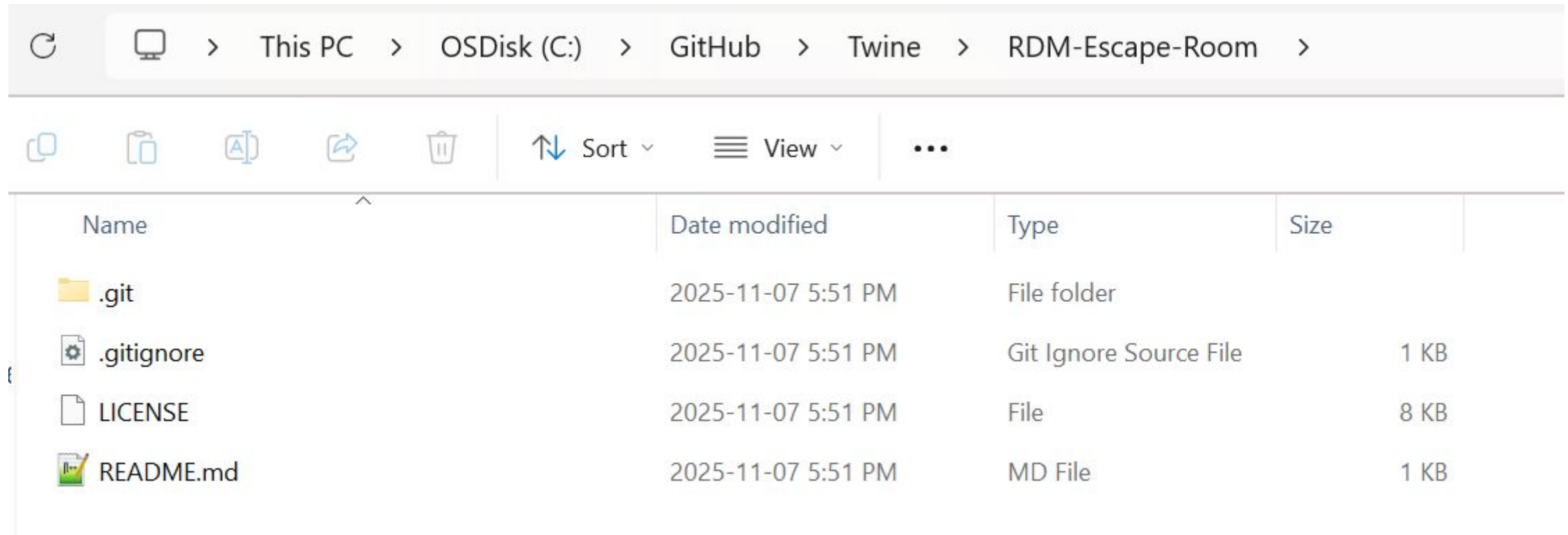
Local path

C:\GitHub\Twine\RDM-Escape-Room Choose...

Clone Cancel

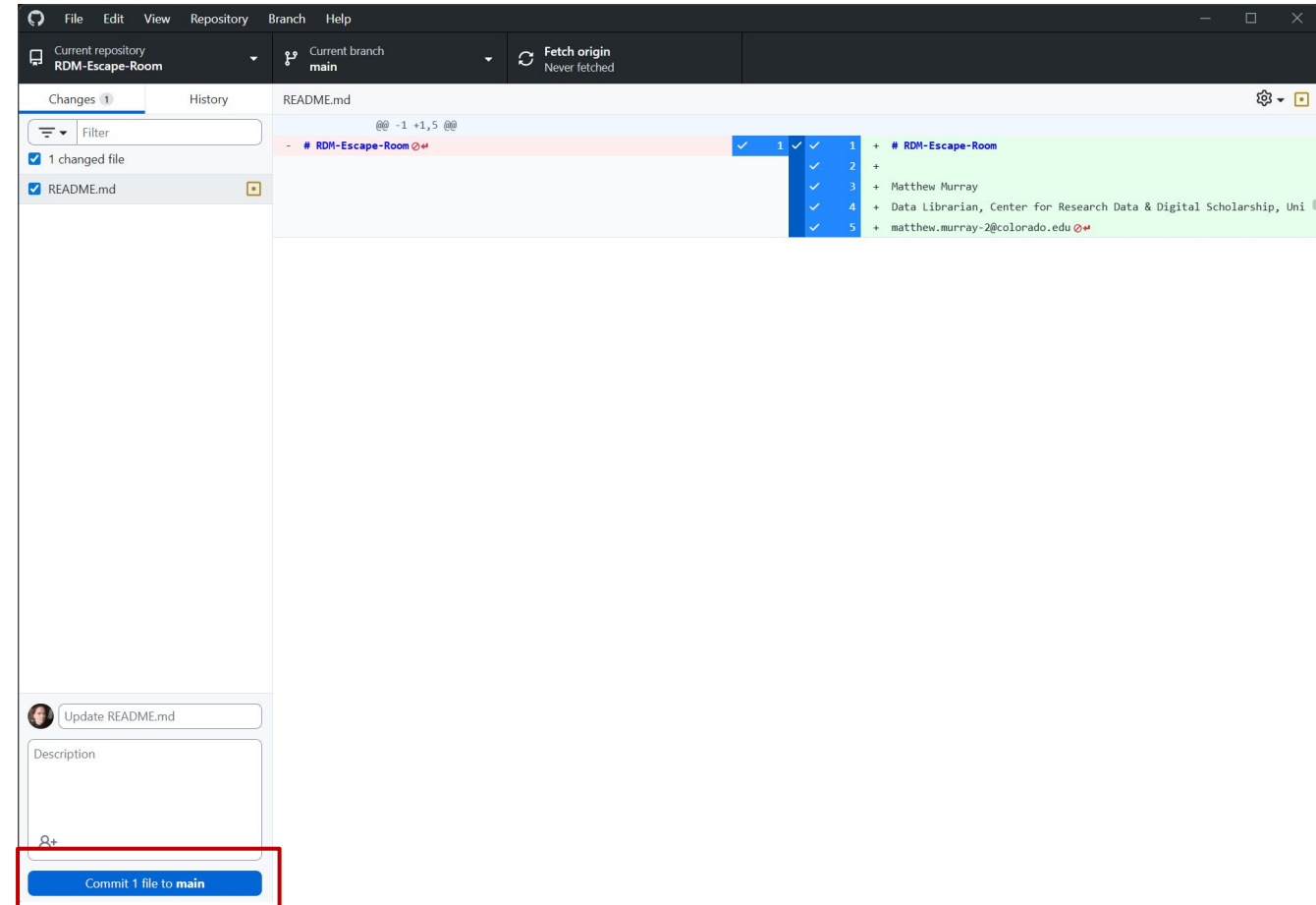
What does this do?

- Copies all files from the GitHub repo to your computer



Updating GitHub Repos

- When you make a change to a file those changes will appear in GitHub
- Click “Commit 1 file to main”
 - Write a description of what changes you made



Updating GitHub Repos

- The commit has been made in Git, but hasn't been pushed to GitHub
- Click “Push origin”

No local changes

There are no uncommitted changes in this repository. Here are some friendly suggestions for what to do next.



Push commits to the origin remote

You have 1 local commit waiting to be pushed to GitHub.

Always available in the toolbar when there are local commits waiting to be pushed or

Ctrl + P

Push origin

Open the repository in your external editor

Select your editor in [Options](#)

Repository menu or Ctrl + Shift + A

Open in Notepad++

View the files of your repository in Explorer

Repository menu or Ctrl + Shift + F

Show in Explorer

Open the repository page on GitHub in your browser

Repository menu or Ctrl + Shift + G

View on GitHub

main1 Branch 0 TagsGo to fileAdd fileCodemidnitelibrary Update README.md419fa29 · now4 Commits.gitignoreCreate .gitignore8 minutes agoLICENSEInitial commit9 minutes agoREADME.mdUpdate README.mdnowREADMECC0-1.0 license

RDM-Escape-Room

Matthew Murray

Data Librarian, Center for Research Data & Digital Scholarship, University of Colorado Boulder Libraries

matthew.murray-2@colorado.edu